

**LUXEMBOURG AP, Luxembourg**WMO: **065900**Lat: **49.627N**Lon: **6.212E**Elev: **1234**StdP: **14.05**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 18.2       | 21.5 | 9.2                           | 9.2 | 23.8 | 14.0 | 11.7 | 25.5 | 27.5                  | 46.3 | 24.5 | 45.1 | 6.8                   | 50   | 0.538 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.5                   | 85.7            | 66.2 | 82.0 | 64.8 | 78.7 | 63.5 | 68.0                | 81.3 | 66.5 | 78.6 | 65.1 | 76.0 | 6.9                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.9                            | 93.0 | 71.9 | 62.3 | 87.8 | 70.3 | 60.9 | 83.5 | 69.6 | 33.0          | 81.6 | 31.8 | 79.2 | 30.6 | 76.1 | 74.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |
| Min               | Max  |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |
| 21.0              | 18.3 | 16.1 | 13.0                       | 91.6 | 4.8                | 3.8 | 9.6                                                | 94.4 | 6.7        | 96.6 | 4.0        | 98.7 | 0.5        | 101.5 |
|                   |      |      | 12.1                       | 70.9 | 5.0                | 1.7 | 8.5                                                | 72.1 | 5.5        | 73.1 | 2.7        | 74.0 | -0.9       | 75.2  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual       | Jan          | Feb          | Mar          | Apr          | May          | Jun          | Jul          | Aug          | Sep          | Oct          | Nov          | Dec          |      |      |
|-------------------------------------------------------------------------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------|------|
|                                                                               |             | (d)          | (e)          | (f)          | (g)          | (h)          | (i)          | (j)          | (k)          | (l)          | (m)          | (n)          | (o)          | (p)          |      |      |
| (6) <b>Temperatures, Degree-Days and Degree-Hours</b>                         | DBAvg       | <b>50.0</b>  | <b>34.4</b>  | <b>36.4</b>  | <b>42.6</b>  | <b>49.4</b>  | <b>56.4</b>  | <b>62.4</b>  | <b>66.1</b>  | <b>65.2</b>  | <b>58.2</b>  | <b>50.6</b>  | <b>41.7</b>  | <b>36.1</b>  | (6)  | (7)  |
|                                                                               | DBStd       | <b>13.07</b> | <b>7.26</b>  | <b>7.08</b>  | <b>6.92</b>  | <b>7.44</b>  | <b>6.92</b>  | <b>6.80</b>  | <b>6.71</b>  | <b>6.50</b>  | <b>5.69</b>  | <b>6.32</b>  | <b>6.35</b>  | <b>7.18</b>  | (7)  | (8)  |
|                                                                               | HDD50       | <b>1991</b>  | <b>484</b>   | <b>382</b>   | <b>245</b>   | <b>100</b>   | <b>18</b>    | <b>1</b>     | <b>0</b>     | <b>0</b>     | <b>3</b>     | <b>70</b>    | <b>256</b>   | <b>432</b>   | (8)  | (9)  |
|                                                                               | HDD65       | <b>5729</b>  | <b>949</b>   | <b>801</b>   | <b>695</b>   | <b>470</b>   | <b>278</b>   | <b>131</b>   | <b>70</b>    | <b>77</b>    | <b>217</b>   | <b>446</b>   | <b>698</b>   | <b>897</b>   | (9)  | (10) |
|                                                                               | CDD50       | <b>2001</b>  | <b>0</b>     | <b>0</b>     | <b>15</b>    | <b>82</b>    | <b>217</b>   | <b>371</b>   | <b>499</b>   | <b>470</b>   | <b>249</b>   | <b>89</b>    | <b>8</b>     | <b>1</b>     | (10) | (11) |
|                                                                               | CDD65       | <b>264</b>   | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>2</b>     | <b>13</b>    | <b>51</b>    | <b>103</b>   | <b>82</b>    | <b>13</b>    | <b>0</b>     | <b>0</b>     | <b>0</b>     | (11) | (12) |
|                                                                               | CDH74       | <b>2026</b>  | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>11</b>    | <b>86</b>    | <b>391</b>   | <b>818</b>   | <b>642</b>   | <b>77</b>    | <b>1</b>     | <b>0</b>     | <b>0</b>     | (12) | (13) |
|                                                                               | CDH80       | <b>553</b>   | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>11</b>    | <b>93</b>    | <b>235</b>   | <b>207</b>   | <b>7</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | (13) | (14) |
| (14) <b>Wind</b>                                                              | WSAvg       | <b>7.4</b>   | <b>8.3</b>   | <b>8.2</b>   | <b>8.5</b>   | <b>7.7</b>   | <b>7.3</b>   | <b>6.9</b>   | <b>6.8</b>   | <b>6.3</b>   | <b>6.8</b>   | <b>7.0</b>   | <b>7.0</b>   | <b>8.2</b>   | (14) | (15) |
| (15) <b>Precipitation</b>                                                     | PrecAvg     | <b>32.8</b>  | <b>2.8</b>   | <b>2.3</b>   | <b>2.4</b>   | <b>2.1</b>   | <b>3.0</b>   | <b>2.9</b>   | <b>2.8</b>   | <b>2.9</b>   | <b>2.6</b>   | <b>2.9</b>   | <b>3.1</b>   | <b>3.3</b>   | (15) | (16) |
|                                                                               | PrecMax     | <b>47.2</b>  | <b>7.8</b>   | <b>6.1</b>   | <b>5.4</b>   | <b>6.9</b>   | <b>7.5</b>   | <b>7.5</b>   | <b>7.8</b>   | <b>7.9</b>   | <b>5.7</b>   | <b>8.0</b>   | <b>6.1</b>   | <b>9.1</b>   | (16) | (17) |
|                                                                               | PrecMin     | <b>20.9</b>  | <b>0.2</b>   | <b>0.2</b>   | <b>0.3</b>   | <b>0.2</b>   | <b>0.3</b>   | <b>0.2</b>   | <b>0.2</b>   | <b>0.6</b>   | <b>0.6</b>   | <b>0.1</b>   | <b>0.4</b>   | <b>0.2</b>   | (17) | (18) |
|                                                                               | PrecStd     | <b>6.4</b>   | <b>1.6</b>   | <b>1.5</b>   | <b>1.3</b>   | <b>1.4</b>   | <b>1.6</b>   | <b>1.7</b>   | <b>1.5</b>   | <b>1.6</b>   | <b>1.4</b>   | <b>1.8</b>   | <b>1.3</b>   | <b>1.9</b>   | (18) | (19) |
| (19) <b>Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures</b> | 0.4%        | DB           | <b>52.0</b>  | <b>55.1</b>  | <b>65.5</b>  | <b>75.7</b>  | <b>81.2</b>  | <b>87.7</b>  | <b>91.3</b>  | <b>92.6</b>  | <b>80.7</b>  | <b>70.7</b>  | <b>59.0</b>  | <b>52.9</b>  | (19) | (20) |
|                                                                               |             | MCWB         | <b>49.6</b>  | <b>46.6</b>  | <b>50.8</b>  | <b>57.6</b>  | <b>64.5</b>  | <b>68.1</b>  | <b>67.7</b>  | <b>68.1</b>  | <b>63.8</b>  | <b>60.8</b>  | <b>55.0</b>  | <b>50.3</b>  | (20) | (21) |
|                                                                               | 2%          | DB           | <b>49.6</b>  | <b>50.9</b>  | <b>60.5</b>  | <b>70.3</b>  | <b>76.0</b>  | <b>82.5</b>  | <b>85.7</b>  | <b>85.3</b>  | <b>76.0</b>  | <b>65.9</b>  | <b>55.2</b>  | <b>50.6</b>  | (21) | (22) |
|                                                                               |             | MCWB         | <b>47.6</b>  | <b>45.4</b>  | <b>48.9</b>  | <b>54.9</b>  | <b>60.5</b>  | <b>65.6</b>  | <b>66.1</b>  | <b>65.9</b>  | <b>62.4</b>  | <b>58.2</b>  | <b>52.1</b>  | <b>48.5</b>  | (22) | (23) |
|                                                                               | 5%          | DB           | <b>46.7</b>  | <b>48.1</b>  | <b>56.3</b>  | <b>65.8</b>  | <b>72.4</b>  | <b>78.5</b>  | <b>82.1</b>  | <b>80.8</b>  | <b>71.6</b>  | <b>62.2</b>  | <b>52.7</b>  | <b>48.3</b>  | (23) | (24) |
|                                                                               |             | MCWB         | <b>44.8</b>  | <b>44.7</b>  | <b>47.5</b>  | <b>53.1</b>  | <b>58.9</b>  | <b>63.9</b>  | <b>64.8</b>  | <b>64.4</b>  | <b>60.8</b>  | <b>56.9</b>  | <b>50.3</b>  | <b>46.7</b>  | (24) | (25) |
|                                                                               | 10%         | DB           | <b>44.2</b>  | <b>46.0</b>  | <b>52.6</b>  | <b>61.7</b>  | <b>68.6</b>  | <b>74.7</b>  | <b>78.6</b>  | <b>76.7</b>  | <b>67.8</b>  | <b>59.3</b>  | <b>50.4</b>  | <b>46.0</b>  | (25) | (26) |
|                                                                               |             | MCWB         | <b>42.5</b>  | <b>43.4</b>  | <b>46.3</b>  | <b>51.1</b>  | <b>57.0</b>  | <b>62.1</b>  | <b>63.6</b>  | <b>63.1</b>  | <b>59.1</b>  | <b>55.4</b>  | <b>48.3</b>  | <b>44.5</b>  | (26) | (27) |
| (27) <b>Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures</b> | 0.4%        | WB           | <b>50.4</b>  | <b>49.0</b>  | <b>53.0</b>  | <b>59.1</b>  | <b>66.2</b>  | <b>70.1</b>  | <b>69.8</b>  | <b>69.8</b>  | <b>65.9</b>  | <b>61.9</b>  | <b>55.5</b>  | <b>51.2</b>  | (27) | (28) |
|                                                                               |             | MCDB         | <b>51.6</b>  | <b>51.9</b>  | <b>60.6</b>  | <b>71.8</b>  | <b>77.5</b>  | <b>83.8</b>  | <b>84.6</b>  | <b>83.8</b>  | <b>75.8</b>  | <b>67.8</b>  | <b>57.6</b>  | <b>52.2</b>  | (28) | (29) |
|                                                                               | 2%          | WB           | <b>47.7</b>  | <b>46.8</b>  | <b>50.8</b>  | <b>56.3</b>  | <b>63.2</b>  | <b>67.2</b>  | <b>68.0</b>  | <b>67.5</b>  | <b>63.9</b>  | <b>59.7</b>  | <b>52.6</b>  | <b>48.8</b>  | (29) | (30) |
|                                                                               |             | MCDB         | <b>49.4</b>  | <b>49.3</b>  | <b>57.1</b>  | <b>67.7</b>  | <b>72.7</b>  | <b>78.6</b>  | <b>81.2</b>  | <b>81.2</b>  | <b>73.0</b>  | <b>64.3</b>  | <b>54.6</b>  | <b>50.3</b>  | (30) | (31) |
|                                                                               | 5%          | WB           | <b>45.0</b>  | <b>45.1</b>  | <b>49.0</b>  | <b>54.0</b>  | <b>60.3</b>  | <b>65.3</b>  | <b>66.4</b>  | <b>65.9</b>  | <b>62.0</b>  | <b>57.7</b>  | <b>50.6</b>  | <b>46.8</b>  | (31) | (32) |
|                                                                               |             | MCDB         | <b>46.4</b>  | <b>47.6</b>  | <b>54.4</b>  | <b>63.4</b>  | <b>69.2</b>  | <b>75.7</b>  | <b>79.3</b>  | <b>77.5</b>  | <b>69.7</b>  | <b>61.5</b>  | <b>52.4</b>  | <b>48.1</b>  | (32) | (33) |
|                                                                               | 10%         | WB           | <b>42.7</b>  | <b>43.4</b>  | <b>47.1</b>  | <b>52.0</b>  | <b>58.2</b>  | <b>63.3</b>  | <b>64.8</b>  | <b>64.3</b>  | <b>60.1</b>  | <b>55.8</b>  | <b>48.5</b>  | <b>44.7</b>  | (33) | (34) |
|                                                                               |             | MCDB         | <b>44.0</b>  | <b>45.8</b>  | <b>51.8</b>  | <b>60.2</b>  | <b>66.6</b>  | <b>72.6</b>  | <b>76.2</b>  | <b>74.4</b>  | <b>66.6</b>  | <b>58.9</b>  | <b>50.3</b>  | <b>46.0</b>  | (34) | (35) |
| (35) <b>Mean Daily Temperature Range</b>                                      | 5% DB       | MDBR         | <b>7.4</b>   | <b>9.5</b>   | <b>12.7</b>  | <b>15.0</b>  | <b>15.7</b>  | <b>16.4</b>  | <b>16.5</b>  | <b>15.9</b>  | <b>14.2</b>  | <b>11.3</b>  | <b>7.8</b>   | <b>7.1</b>   | (35) | (36) |
|                                                                               |             | MCDBR        | <b>9.4</b>   | <b>12.4</b>  | <b>18.3</b>  | <b>20.2</b>  | <b>20.4</b>  | <b>21.1</b>  | <b>21.4</b>  | <b>21.8</b>  | <b>19.4</b>  | <b>15.0</b>  | <b>9.9</b>   | <b>8.2</b>   | (36) | (37) |
|                                                                               | 5% WB       | MCWBR        | <b>8.5</b>   | <b>8.8</b>   | <b>10.1</b>  | <b>9.8</b>   | <b>9.8</b>   | <b>9.7</b>   | <b>8.3</b>   | <b>8.7</b>   | <b>9.1</b>   | <b>8.9</b>   | <b>7.6</b>   | <b>7.4</b>   | (37) | (38) |
|                                                                               |             | MCWBR        | <b>9.2</b>   | <b>10.6</b>  | <b>14.7</b>  | <b>18.3</b>  | <b>18.5</b>  | <b>19.1</b>  | <b>19.4</b>  | <b>18.6</b>  | <b>16.8</b>  | <b>13.1</b>  | <b>9.3</b>   | <b>8.1</b>   | (38) | (39) |
| (39) <b>Clear-Sky Solar Irradiance</b>                                        | taub        |              | <b>0.301</b> | <b>0.327</b> | <b>0.355</b> | <b>0.385</b> | <b>0.395</b> | <b>0.397</b> | <b>0.393</b> | <b>0.390</b> | <b>0.363</b> | <b>0.347</b> | <b>0.321</b> | <b>0.300</b> | (39) | (40) |
|                                                                               |             | taud         | <b>2.447</b> | <b>2.407</b> | <b>2.340</b> | <b>2.280</b> | <b>2.281</b> | <b>2.299</b> | <b>2.300</b> | <b>2.339</b> | <b>2.395</b> | <b>2.437</b> | <b>2.452</b> | <b>2.438</b> | (40) | (41) |
|                                                                               | Ebn at Noon |              | <b>239</b>   | <b>259</b>   | <b>270</b>   | <b>272</b>   | <b>274</b>   | <b>273</b>   | <b>272</b>   | <b>268</b>   | <b>264</b>   | <b>247</b>   | <b>228</b>   | <b>222</b>   | (41) | (42) |
|                                                                               |             | Edh at Noon  | <b>20</b>    | <b>26</b>    | <b>33</b>    | <b>38</b>    | <b>40</b>    | <b>40</b>    | <b>39</b>    | <b>36</b>    | <b>31</b>    | <b>25</b>    | <b>20</b>    | <b>17</b>    | (42) | (43) |
| (43) <b>All-Sky Solar Radiation</b>                                           | RadAvg      |              | <b>255</b>   | <b>487</b>   | <b>865</b>   | <b>1352</b>  | <b>1597</b>  | <b>1777</b>  | <b>1687</b>  | <b>1443</b>  | <b>1089</b>  | <b>626</b>   | <b>289</b>   | <b>201</b>   | (43) | (44) |
|                                                                               | RadStd      |              | <b>39</b>    | <b>90</b>    | <b>128</b>   | <b>170</b>   | <b>169</b>   | <b>159</b>   | <b>173</b>   | <b>153</b>   | <b>125</b>   | <b>57</b>    | <b>32</b>    | <b>31</b>    | (44) | (45) |

**Histor**